SUPERSINGULAR CURVES AND THE TRACE OF FROBENIUS: SOLUTIONS

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Section 1 proves that an elliptic curve over $\mathbb{F}_q$ is supersingular if and only if p divides its trace of Frobenius, and works out what this says over $\mathbb{F}_p$, over $\mathbb{F}_2$ and $\mathbb{F}_3$, and over $\mathbb{F}_8$. Section 2 shows that $E(\mathbb{F}_q) \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/mn\mathbb{Z}$ with $m \mid q-1$, and pins down m for a supersingular curve over $\mathbb{F}_p$. Section 3 shows that an elliptic curve over $\mathbb{Q}$ is ordinary modulo infinitely many primes. Each problem is preceded by the definitions and results it uses.

Reading. [Sil09, §§V.3–V.4], with [Sil09, §V.2] for the trace of Frobenius, [Sil09, §III.6] for the torsion subgroups and [Sil09, §III.8] for the Weil pairing. The three problems are exercises from [Sil09, Chapter V]: Problem 1.12 is Exercise 5.10, Problem 2.3 is Exercise 5.6 and Problem 3.8 is Exercise 5.11.

`Magma`**.** If you have no local installation, every block below runs in the free online calculator.

Standing notation.

- A power $q = p^e$ of a prime p.
- The p^r-power Frobenius morphism $\mathsf{F}^r \colon C \to C^{(p^r)}$ of a curve C over a field of characteristic p, for $r \geqslant 1$, which raises coordinates to the p^r-th power. For an elliptic curve E, its dual isogeny $\mathsf{V}^r \colon E^{(p^r)} \to E$ is the Verschiebung.
- The q-power Frobenius *endomorphism* of an elliptic curve E over $\mathbb{F}_q$. Since $E^{(q)} = E$, the morphism
$$\mathsf{F}^e \colon E \to E, \qquad (x, y) \mapsto (x^q, y^q),$$
is an endomorphism of E, with dual $\mathsf{V}^e \in \mathrm{End}(E)$. Here $\mathrm{End}(E)$ consists of the endomorphisms defined over $\mathbb{F}_q$.
- The group $E[m](\overline{\mathbb{F}}_q)$ of m-torsion points of E with coordinates in $\overline{\mathbb{F}}_q$, for $m \geqslant 1$.
- The supersingular elliptic curves E over $\mathbb{F}_q$: those with $E[p^r](\overline{\mathbb{F}}_q) = 0$ for one, equivalently every, $r \geqslant 1$ [Sil09, Theorem V.3.1 and the definition after it]. The others are ordinary. Silverman also says that a supersingular curve has "Hasse invariant 0" and an ordinary one "Hasse invariant 1". That is the phrasing of Exercise 5.11.

How to use this. Allow about 35 minutes for §1, 20 for §2 and 25 for §3. In Problem 1.12 the heart is parts (1) and (5). Parts (3) and (4) are hand counts to finish at home once you have looked at the tables. In Problem 2.3, part (3) is the point. Problem 3.8 is one argument in three steps. Do parts (2) and (3) in the session and part (1) at home. The `Magma` blocks return in a few seconds.

1. Supersingularity and the trace of Frobenius

This section is [Sil09, Exercise 5.10]. Part (5) is a little stronger than [Sil09, Exercise 5.10(e)], and part (6) needs the stronger form. By Deuring's theorem [Sil09, Theorem V.3.1], a supersingular curve is one whose Verschiebung is purely inseparable, whose endomorphism ring over $\overline{\mathbb{F}}_q$ is an order in a quaternion algebra, and whose formal group has height 2. None of these is easy to read off from a Weierstrass equation. Over a finite field there is a criterion that is: count the rational points.

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Definition 1.1 (Trace of Frobenius)**.** Let E be an elliptic curve over $\mathbb{F}_q$. The trace of Frobenius of E is

$$a \coloneqq q + 1 - \#E(\mathbb{F}_q).$$

Definition 1.2 (Separable degree)**.** For an isogeny ψ, $\deg_s \psi$ denotes its separable degree.

In the following results, E is an elliptic curve over $\mathbb{F}_q$, with $q = p^e$, and a is its trace of Frobenius.

Proposition 1.3 (Frobenius is purely inseparable [Sil09, Proposition II.2.11])**.** *For $r \geqslant 1$, the Frobenius morphism F^r is purely inseparable of degree p^r.*

Corollary 1.4 (Purely inseparable isogenies [Sil09, Corollary II.2.12])**.** *A purely inseparable isogeny $\psi\colon E \to E'$ of degree p^r factors as $\psi = \lambda \circ \mathsf{F}^r$ with $\lambda\colon E^{(p^r)} \to E'$ an isomorphism.*

Theorem 1.5 (Kernels and separable degree [Sil09, Theorem III.4.10(a)])**.** *A nonzero isogeny ψ satisfies $\#\ker\psi = \deg_s \psi$, the kernel taken on points with coordinates in $\overline{\mathbb{F}}_q$.*

Corollary 1.6 (Separability of $[m] + [n]\mathsf{F}^e$ [Sil09, Corollary III.5.5])**.** *For $m, n \in \mathbb{Z}$, the endomorphism $[m] + [n]\mathsf{F}^e$ is separable if and only if $p \nmid m$.*

Theorem 1.7 (Dual isogenies [Sil09, Theorem III.6.2(a)])**.** *Every isogeny ψ satisfies $\hat{\psi} \circ \psi = [\deg\psi]$ and $\psi \circ \hat{\psi} = [\deg\psi]$. So $\mathsf{V}^e \circ \mathsf{F}^e = \mathsf{F}^e \circ \mathsf{V}^e = [q]$.*

Corollary 1.8 (The degree of $[m]$ [Sil09, Corollary III.6.4(a)])**.** *For $m \in \mathbb{Z}$, the multiplication-by-m map has degree $\deg[m] = m^2$.*

Corollary 1.9 (The p-power torsion [Sil09, Corollary III.6.4(c)])**.** *Either $E[p^r](\overline{\mathbb{F}}_q) = 0$ for every $r \geqslant 1$, or $E[p^r](\overline{\mathbb{F}}_q) \cong \mathbb{Z}/p^r\mathbb{Z}$ for every $r \geqslant 1$.*

Theorem 1.10 (The characteristic polynomial of Frobenius [Sil09, Theorem V.2.3.1(b)])**.** *The Frobenius endomorphism satisfies $(\mathsf{F}^e)^2 - [a]\mathsf{F}^e + [q] = 0$ in $\mathrm{End}(E)$.*

Theorem 1.11 (Hasse's bound [Sil09, Theorem V.1.1])**.** *The trace of Frobenius satisfies $|a| \leqslant 2\sqrt{q}$.*

Problem 1.12 (Supersingularity and the trace of Frobenius)**.**
Let E be an elliptic curve over $\mathbb{F}_q$.
(1) Show that $\mathsf{V}^e = [a] - \mathsf{F}^e$ in $\mathrm{End}(E)$, and deduce that V^e is separable if and only if $p \nmid a$. Show that $\ker\mathsf{V}^e$ has $\#E[q](\overline{\mathbb{F}}_q)$ elements, so that $\deg_s \mathsf{V}^e \in \{1, q\}$. Conclude that

$$E \text{ is supersingular} \iff a \equiv 0 \pmod{p}.$$

Solution. By Theorems 1.7 and 1.10,

$$0 = (\mathsf{F}^e)^2 - [a]\mathsf{F}^e + \mathsf{V}^e \circ \mathsf{F}^e = \left(\mathsf{F}^e - [a] + \mathsf{V}^e\right) \circ \mathsf{F}^e.$$

The map F^e is surjective on points with coordinates in $\overline{\mathbb{F}}_q$, since $(x, y) = \mathsf{F}^e(x^{1/q}, y^{1/q})$. So the left factor vanishes on all of $E(\overline{\mathbb{F}}_q)$, that is, $\mathsf{V}^e = [a] - \mathsf{F}^e$. By Corollary 1.6 with $m = a$ and $n = -1$, V^e is separable if and only if $p \nmid a$.
For the kernel: $\mathsf{V}^e \circ \mathsf{F}^e = [q]$ maps $E[q](\overline{\mathbb{F}}_q)$ to O, so F^e maps $E[q](\overline{\mathbb{F}}_q)$ into $\ker\mathsf{V}^e$. Conversely a point Q of $E(\overline{\mathbb{F}}_q)$ in $\ker\mathsf{V}^e$ is $\mathsf{F}^e(P)$ for some $P \in E(\overline{\mathbb{F}}_q)$, and then $[q]P = O$. So $\ker\mathsf{V}^e = \mathsf{F}^e\big(E[q](\overline{\mathbb{F}}_q)\big)$, and F^e is injective on $E(\overline{\mathbb{F}}_q)$, because $t \mapsto t^q$ is injective on the field $\overline{\mathbb{F}}_q$. Hence $\#\ker\mathsf{V}^e = \#E[q](\overline{\mathbb{F}}_q)$, which is 1 or q by Corollary 1.9 with $r = e$. By Theorem 1.5, $\deg_s \mathsf{V}^e \in \{1, q\}$. And $\deg\mathsf{V}^e = q$, since $\deg\mathsf{V}^e \cdot \deg\mathsf{F}^e = \deg[q] = q^2$ by Theorem 1.7, Corollary 1.8, and Proposition 1.3. So V^e is either separable ($\deg_s = q$) or purely inseparable ($\deg_s = 1$), and

$$E \text{ supersingular} \iff E[q](\overline{\mathbb{F}}_q) = 0 \iff \deg_s \mathsf{V}^e = 1 \iff \mathsf{V}^e \text{ not separable} \iff p \mid a.$$

The first equivalence is the definition, with $r = e$. The last is the first paragraph.

(2) Suppose $q = p \geqslant 5$. Show that E is supersingular if and only if $\#E(\mathbb{F}_p) = p + 1$. Where is $p \geqslant 5$ used?

SOLUTION. **To make $a = 0$ the only multiple of p in the Hasse interval.** For $q = p$, part (1) says that E is supersingular if and only if $p \mid a$. By Theorem 1.11, $|a| \leqslant 2\sqrt{p}$, and $2\sqrt{p} < p$ exactly when $p > 4$. So for $p \geqslant 5$ the only multiple of p in the Hasse interval is $a = 0$, that is, $\#E(\mathbb{F}_p) = p + 1$. For $p = 2$ and $p = 3$ the interval also contains $a = \pm p$, and parts (3) and (4) show that these values occur.

(3) Every elliptic curve over $\mathbb{F}_3$ is isomorphic to one given by $y^2 = x^3 + a_2x^2 + a_4x + a_6$. `Magma` lists the isomorphism classes, with $\#E(\mathbb{F}_3)$, a and whether E is supersingular:

```
K := GF(3);  reps := [];
for a2, a4, a6 in K do
  ok, E := IsEllipticCurve([K | 0, a2, 0, a4, a6]);
  if ok and not exists{F : F in reps | IsIsomorphic(E, F)} then
    Append(~reps, E);
  end if;
end for;
#reps;                                                          // 8
for E in reps do
  printf "%o  %o  %o  %o\n", aInvariants(E), #E,
         TraceOfFrobenius(E), IsSupersingular(E);
end for;
// [ 0, 0, 0, 1, 0 ]  4  0  true
// [ 0, 0, 0, 2, 0 ]  4  0  true
// [ 0, 0, 0, 2, 1 ]  7  -3  true
// [ 0, 0, 0, 2, 2 ]  1  3  true
// [ 0, 1, 0, 0, 1 ]  6  -2  false
// [ 0, 1, 0, 0, 2 ]  3  1  false
// [ 0, 2, 0, 0, 1 ]  5  -1  false
// [ 0, 2, 0, 0, 2 ]  2  2  false
```

Which column decides supersingularity, by part (1)? Count the rational points of $y^2 = x^3 - x + 1$ and of $y^2 = x^3 - x - 1$ by hand, and explain why part (2) fails for $p = 3$.

SOLUTION. **The third column, a. The two curves have 7 and 1 rational points.** E is supersingular exactly when $3 \mid a$, that is, when $\#E(\mathbb{F}_3) \equiv 1 \pmod 3$. The table shows the four classes with $a_2 = 0$ (these are the curves with $j = 0$) and the four with $a_2 \neq 0$. Over $\mathbb{F}_3$ every x satisfies $x^3 = x$, so $x^3 - x + 1 = 1$ for all $x \in \mathbb{F}_3$: the equation $y^2 = x^3 - x + 1$ has the two solutions $y = \pm 1$ for each of the three values of x, and with O, $\#E(\mathbb{F}_3) = 7$, $a = -3$. Likewise $x^3 - x - 1 = -1 = 2$ is not a square in $\mathbb{F}_3$, so $y^2 = x^3 - x - 1$ has no affine rational points and $\#E(\mathbb{F}_3) = 1$, $a = 3$. Both curves are supersingular with $\#E(\mathbb{F}_3) \neq 4$: part (2) fails for $p = 3$ because $a = \pm 3$ lies in the Hasse interval $|a| \leqslant 2\sqrt{3} \approx 3.46$.

(4) The same over $\mathbb{F}_2$, where a Weierstrass equation needs all five coefficients:

```
K := GF(2);  reps := [];
for a1, a2, a3, a4, a6 in K do
  ok, E := IsEllipticCurve([K | a1, a2, a3, a4, a6]);
  if ok and not exists{F : F in reps | IsIsomorphic(E, F)} then
    Append(~reps, E);
```

```
  end if;
end for;
#reps;                                                          // 5
for E in reps do
  printf "%o  %o  %o  %o\n", aInvariants(E), #E,
         TraceOfFrobenius(E), IsSupersingular(E);
end for;
// [ 0, 0, 1, 0, 0 ]  3  0  true
// [ 0, 0, 1, 1, 0 ]  5  -2  true
// [ 0, 0, 1, 1, 1 ]  1  2  true
// [ 1, 0, 0, 0, 1 ]  4  -1  false
// [ 1, 0, 1, 0, 1 ]  2  1  false
```

Count the rational points of the three supersingular curves by hand. Which coefficient decides supersingularity over $\mathbb{F}_2$, and what does the analogue of part (2) say for $p = 2$?

Solution. **3, 5 and 1 rational points. Supersingular means $a_1 = 0$, not $a = 0$.** Over $\mathbb{F}_2$, E is supersingular exactly when a is even, that is, when $\#E(\mathbb{F}_2)$ is odd. In the table this is the condition $a_1 = 0$. For the three supersingular curves, $y^2 + y$ takes the value 0 at $y = 0, 1$ and never the value 1:

- $y^2 + y = x^3$: two rational points over $x = 0$, none over $x = 1$. With O, $\#E(\mathbb{F}_2) = 3$ and $a = 0$.
- $y^2 + y = x^3 + x$: the right side is 0 for both x, so four affine rational points, $\#E(\mathbb{F}_2) = 5$ and $a = -2$.
- $y^2 + y = x^3 + x + 1$: the right side is 1 for both x, so no affine rational points, $\#E(\mathbb{F}_2) = 1$ and $a = 2$.

The analogue of part (2) would say $\#E(\mathbb{F}_2) = 3$, and two of the three supersingular curves violate it: $a = \pm 2$ lies in the Hasse interval $|a| \leqslant 2\sqrt{2} \approx 2.83$.

Remark. Again $j = 0$. Compare [Sil09, Exercise 5.7].

(5) Suppose $p \mid a$. Use Corollary 1.4 to write $\mathsf{V}^e = \lambda \circ \mathsf{F}^e$ with $\lambda \in \mathrm{Aut}(E)$, deduce that $[a] = ([1] + \lambda) \circ \mathsf{F}^e$, and conclude that $q \mid a^2$. So

$$a \equiv 0 \pmod{p} \iff q \mid a^2 \iff p^{\lceil e/2 \rceil} \mid a.$$

Solution. If $p \mid a$, then by part (1) V^e is purely inseparable, of degree $q = p^e$. By Corollary 1.4, $\mathsf{V}^e = \lambda \circ \mathsf{F}^e$ with $\lambda \colon E^{(q)} \to E$ an isomorphism. Since $E^{(q)} = E$, $\lambda \in \mathrm{Aut}(E)$. Then

$$[a] = \mathsf{F}^e + \mathsf{V}^e = \mathsf{F}^e + \lambda \circ \mathsf{F}^e = ([1] + \lambda) \circ \mathsf{F}^e,$$

and taking degrees, which are multiplicative, $a^2 = \deg([1] + \lambda) \cdot q$ by Corollary 1.8 and Proposition 1.3. If $[1] + \lambda = 0$ this reads $a = 0$. So $q \mid a^2$. Conversely $q \mid a^2$ implies $p \mid a$. Finally, writing $a = p^k b$ with $p \nmid b$, $p^e \mid a^2 = p^{2k} b^2$ if and only if $2k \geqslant e$, that is, $k \geqslant \lceil e/2 \rceil$. For $q = p^{2i}$ this is $p^i \mid a$. For $q = p^{2i+1}$ it is $p^{i+1} \mid a$.

(6) Show that no elliptic curve over $\mathbb{F}_8$ has exactly 7 or exactly 11 rational points. Magma collects the values of $\#E(\mathbb{F}_8)$ over all Weierstrass equations, and the corresponding traces:

```
K := GF(8);  counts := {};
for a1, a2, a3, a4, a6 in K do
  ok, E := IsEllipticCurve([a1, a2, a3, a4, a6]);
  if ok then Include(~counts, #E); end if;
end for;
```

```
counts;                          // { 4, 5, 6, 8, 9, 10, 12, 13, 14 }
{ 9 - n : n in counts };         // { -5, -4, -3, -1, 0, 1, 3, 4, 5 }
```

Which integers in the Hasse interval are missing, and which traces belong to supersingular curves? Predict both from parts (1) and (5) before you look.

SOLUTION. **Missing: $a = \pm 2$. Supersingular: $a \in \{0, \pm 4\}$.** Over $\mathbb{F}_8$, $\#E(\mathbb{F}_8) = 7$ or 11 means $a = 2$ or $a = -2$. Then $2 \mid a$, so by part (5) $8 \mid a^2 = 4$, which is false. The Hasse interval is $|a| \leqslant 2\sqrt{8} \approx 5.66$, so $a \in [-5, 5]$ and $\#E(\mathbb{F}_8) \in [4, 14]$. The values 7 and 11 are the only ones missing from the `Magma` set. The supersingular traces are the even a with $8 \mid a^2$, that is, $4 \mid a$: $a \in \{0, \pm 4\}$, and $\#E(\mathbb{F}_8) \in \{5, 9, 13\}$. All three occur.

Remark 1.13 (The traces of supersingular curves)**.** The identity $a^2 = \deg([1] + \lambda) \cdot q$ of Problem 1.12(5) says more than $q \mid a^2$. Since $a^2 \leqslant 4q$, the integer $d := \deg([1] + \lambda)$ is at most 4, so a supersingular curve over $\mathbb{F}_q$ has

$$a^2 \in \{0,\ q,\ 2q,\ 3q,\ 4q\}\,,$$

and only the values with dq a perfect square can occur. Over $\mathbb{F}_8$ these are $a^2 \in \{0, 16\}$, as found above; over $\mathbb{F}_p$ with $p \geqslant 5$ only $a = 0$; over $\mathbb{F}_3$ also $a^2 = 9 = 3q$, and over $\mathbb{F}_2$ also $a^2 = 4 = 2q$, the cases of parts (3) and (4). Which of the five values occur, over which q, is a theorem of Waterhouse. We do not need it.

2. The group structure of $E(\mathbb{F}_q)$

This section is [Sil09, Exercise 5.6].

In the following results, E is an elliptic curve over a field K and $m \geqslant 1$ is an integer with $\operatorname{char} K \nmid m$.

Corollary 2.1 (The m-torsion points [Sil09, Corollary III.6.4(b)])**.** *The group $E[m](\overline{K})$ is isomorphic to $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$.*

Corollary 2.2 (Rational torsion and roots of unity [Sil09, Corollary III.8.1.1])**.** *If every m-torsion point of $E(\overline{K})$ is K-rational, then K contains the m-th roots of unity.*

Problem 2.3 (The group structure of $E(\mathbb{F}_q)$)**.**
Let E be an elliptic curve over $\mathbb{F}_q$.

(1) Show that there are integers $m, n \geqslant 1$ with $p \nmid m$ and $E(\mathbb{F}_q) \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/mn\mathbb{Z}$.

SOLUTION. Put $G := E(\mathbb{F}_q)$, a finite abelian group, and $N := \#G$. Then G is the direct sum of its ℓ-primary parts G_ℓ over the primes $\ell \mid N$, and $G_\ell \cong \bigoplus_i \mathbb{Z}/\ell^{k_i}\mathbb{Z}$ for some exponents $k_1 \leqslant \ldots \leqslant k_s$, where the number s of cyclic factors is the dimension of the $\mathbb{F}_\ell$-vector space $G[\ell] = G_\ell[\ell]$. Now $G[\ell]$ is a subgroup of $E[\ell](\overline{\mathbb{F}}_q)$, which is $(\mathbb{Z}/\ell\mathbb{Z})^2$ for $\ell \neq p$ by Corollary 2.1, and 0 or $\mathbb{Z}/p\mathbb{Z}$ for $\ell = p$ by Corollary 1.9. So $s \leqslant 2$ for $\ell \neq p$, and $s \leqslant 1$ for $\ell = p$: write

$$G_\ell \cong \mathbb{Z}/\ell^{i_\ell}\mathbb{Z} \times \mathbb{Z}/\ell^{j_\ell}\mathbb{Z}$$

with $0 \leqslant i_\ell \leqslant j_\ell$ and $i_p = 0$. Put

$$m := \prod_\ell \ell^{i_\ell}, \qquad mn := \prod_\ell \ell^{j_\ell},$$

so that $m \mid mn$ and $p \nmid m$. By the Chinese remainder theorem, $G \cong \bigoplus_\ell G_\ell \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/mn\mathbb{Z}$.

(2) With m as in part (1), show that $q \equiv 1 \pmod{m}$.

SOLUTION. Since $m \mid m$ and $m \mid mn$, the subgroup $G[m]$ of $G \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/mn\mathbb{Z}$ is $(\mathbb{Z}/m\mathbb{Z})^2$, with m^2 elements. It is contained in $E[m](\overline{\mathbb{F}}_q)$, which also has m^2 elements because $p \nmid m$ (Corollary 2.1). So every m-torsion point of $E(\overline{\mathbb{F}}_q)$ is $\mathbb{F}_q$-rational, and by Corollary 2.2 the group $\mathbb{F}_q^\times$ contains the m-th roots of unity, which form a cyclic group of order m (as $p \nmid m$). Since $\mathbb{F}_q^\times$ has order $q-1$, $m \mid q-1$.

(3) Suppose $q = p \geqslant 5$ and E is supersingular. Show that $m = 1$ or $m = 2$, and that $m = 1$ if $p \equiv 1 \pmod 4$.

SOLUTION. By Problem 1.12(2), $\#E(\mathbb{F}_p) = p+1$, so $m^2 n = p+1$ and in particular $m^2 \mid p+1$. By part (2), $m \mid p-1$. Hence m divides $\gcd(p-1, p+1) = 2$, so $m \in \{1, 2\}$. If $m = 2$, then $4 = m^2$ divides $p+1$, so $p \equiv 3 \pmod 4$. Therefore $p \equiv 1 \pmod 4$ forces $m = 1$, and $E(\mathbb{F}_p)$ is cyclic of order $p+1$.

(4) Both values of m occur when $p \equiv 3 \pmod 4$. Over $\mathbb{F}_7$ the curves $y^2 = x^3 + x$ and $y^2 = x^3 - x$, both with $j = 1728$, are supersingular. Here `Invariants(AbelianGroup(E))` returns $[m, mn]$, or $[mn]$ when $m = 1$:

```
E1 := EllipticCurve([GF(7) | 1, 0]);  E2 := EllipticCurve([GF(7) | -1, 0]);
IsSupersingular(E1), IsSupersingular(E2);                    // true true
Invariants(AbelianGroup(E1));                                // [ 8 ]
Invariants(AbelianGroup(E2));                                // [ 2, 4 ]
[ j : j in GF(13) | IsSupersingular(EllipticCurveFromjInvariant(j)) ];  // [ 5 ]
E3 := EllipticCurveFromjInvariant(GF(13)!5);  aInvariants(E3);   // [ 1, 0, 0, 7, 2 ]
Invariants(AbelianGroup(E3));                                // [ 14 ]
Invariants(AbelianGroup(QuadraticTwist(E3)));                // [ 14 ]
```

Explain the two values of m over $\mathbb{F}_7$ from the rational points of order 2 of each curve, and the value over $\mathbb{F}_{13}$ from part (3).

SOLUTION. **$m = 1$ for E_1 and $m = 2$ for E_2 over $\mathbb{F}_7$. Over $\mathbb{F}_{13}$, $m = 1$.** Both curves over $\mathbb{F}_7$ have $j = 1728$ and $7 \equiv 3 \pmod 4$, and `Magma` confirms that they are supersingular, with 8 rational points each. The rational points of order 2 are $(x_0, 0)$ with $x_0 \in \mathbb{F}_7$ a root of the cubic. For $y^2 = x^3 + x = x(x^2+1)$: since $7 \equiv 3 \pmod 4$, -1 is not a square in $\mathbb{F}_7$, so x^2+1 is irreducible and $x = 0$ is the only root. So $E_1(\mathbb{F}_7)$ has a single rational point of order 2, so $E_1(\mathbb{F}_7)[2] \neq (\mathbb{Z}/2\mathbb{Z})^2$, $m = 1$, and $E_1(\mathbb{F}_7) \cong \mathbb{Z}/8\mathbb{Z}$. For $y^2 = x^3 - x = x(x-1)(x+1)$ all three roots are rational, so $E_2(\mathbb{F}_7)[2] = (\mathbb{Z}/2\mathbb{Z})^2$ and $m = 2$: $E_2(\mathbb{F}_7) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$. Over $\mathbb{F}_{13}$, with $13 \equiv 1 \pmod 4$, the only supersingular j-invariant is $j = 5$, and part (3) predicts $m = 1$ for every supersingular curve, which is what `Magma` finds: both curves with $j = 5$ (the second is the quadratic twist of the first) have $E(\mathbb{F}_{13}) \cong \mathbb{Z}/14\mathbb{Z}$.

3. Infinitely many ordinary primes

This section is [Sil09, Exercise 5.11], with "ordinary" for "Hasse invariant 1". Silverman's exercise asks for infinitely many ordinary primes. The curve of part (4) and its supersingular primes are [Sil09, Example V.4.6]. The proof uses results that we have not proved.

Definition 3.1 (Reduction modulo p)**.** Let E be an elliptic curve over $\mathbb{Q}$, given by a Weierstrass equation with coefficients in $\mathbb{Z}$ and discriminant $\Delta \in \mathbb{Z}$. For a prime $p \nmid \Delta$, reducing the coefficients modulo p gives a Weierstrass equation over $\mathbb{F}_p$ with nonzero discriminant, hence an elliptic curve E_p over $\mathbb{F}_p$.

Definition 3.2 (Supersingular and ordinary primes)**.** A prime $p \nmid \Delta$ is supersingular or ordinary for E according as E_p is.

Definition 3.3 (Primes that split completely)**.** Let $L/\mathbb{Q}$ be a finite Galois extension. A prime p splits completely in L if $p\mathcal{O}_L$ is a product of $[L:\mathbb{Q}]$ distinct prime ideals of the ring of integers $\mathcal{O}_L$.

In the following results, E, Δ and E_p are as in Definition 3.1, and $L/\mathbb{Q}$ is a finite Galois extension.

Proposition 3.4 (Reduction modulo p [Sil09, Proposition VII.2.1])**.** *Let $p \nmid \Delta$. Every point of $E(\mathbb{Q}_p)$ can be written $[X:Y:Z]$ with $X,Y,Z \in \mathbb{Z}_p$ not all divisible by p, and reducing the coordinates modulo p gives a group homomorphism $E(\mathbb{Q}_p) \to E_p(\mathbb{F}_p)$.*

Proposition 3.5 (Reduction of torsion [Sil09, Proposition VII.3.1(b)])**.** *Let $p \nmid \Delta$ and let $m \geqslant 1$ with $p \nmid m$. The restriction of the reduction homomorphism $E(\mathbb{Q}_p) \to E_p(\mathbb{F}_p)$ to the m-torsion points of $E(\mathbb{Q}_p)$ is injective.*

Proposition 3.6 (Completions at a completely split prime [Neu99])**.** *Let p be a prime that splits completely in L. For each prime $\mathfrak{p}$ of L above p the completion $L_\mathfrak{p}$ is $\mathbb{Q}_p$, so that L embeds in $\mathbb{Q}_p$.*

Theorem 3.7 (Chebotarev density theorem [Neu99])**.** *Infinitely many primes split completely in L: they have density $1/[L:\mathbb{Q}]$ among all primes.*

Problem 3.8 (Infinitely many ordinary primes)**.**
Fix a prime $\ell \geqslant 3$, and let $L \coloneqq \mathbb{Q}(E[\ell])$ be the field generated over $\mathbb{Q}$ by the affine coordinates of the nonzero ℓ-torsion points of $E(\overline{\mathbb{Q}})$.

(1) Show that L is a finite Galois extension of $\mathbb{Q}$, and that L contains the ℓ-th roots of unity.

Solution. The group $E[\ell](\overline{\mathbb{Q}})$ is finite (it has ℓ^2 elements by Corollary 2.1), so the coordinates of its points are algebraic numbers, each with finitely many conjugates, and L is a finite extension of $\mathbb{Q}$. If $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ and P is an ℓ-torsion point, then $\sigma(P)$ is again an ℓ-torsion point: the multiplication-by-ℓ map is given by rational functions with coefficients in $\mathbb{Q}$, so $[\ell]\sigma(P) = \sigma([\ell]P) = O$. Hence σ permutes the finite set of coordinates generating L, and $\sigma(L) = L$. So $L/\mathbb{Q}$ is normal, hence Galois. The ℓ-torsion points of $E(\overline{\mathbb{Q}})$ all have coordinates in L, so Corollary 2.2, applied to E over $K = L$, gives $\mu_\ell \subset L$.

(2) Let $p \nmid \ell\Delta$ be a prime that splits completely in L. Show that $\ell^2 \mid \#E_p(\mathbb{F}_p)$ and that $p \equiv 1 \pmod{\ell}$.

Solution. By Proposition 3.6, choose an embedding $L \hookrightarrow \mathbb{Q}_p$. Every ℓ-torsion point of $E(\overline{\mathbb{Q}})$ has coordinates in L, so $E(\mathbb{Q}_p)$ contains a copy of $E[\ell](\overline{\mathbb{Q}}) \cong (\mathbb{Z}/\ell\mathbb{Z})^2$. By Propositions 3.4 and 3.5 with $m = \ell$, reduction maps this group injectively into $E_p(\mathbb{F}_p)$, so $E_p(\mathbb{F}_p)$ has a subgroup of order ℓ^2, and $\ell^2 \mid \#E_p(\mathbb{F}_p)$.
For the congruence, let $\zeta \in L \subset \mathbb{Q}_p$ be a primitive ℓ-th root of unity. It is a root of $x^\ell - 1$, so it lies in $\mathbb{Z}_p$, and its reduction $\bar{\zeta}$ is a root of $x^\ell - 1$ in $\mathbb{F}_p$. The ℓ roots $1, \zeta, \dots, \zeta^{\ell-1}$ reduce to ℓ distinct elements of $\mathbb{F}_p$: their reductions are the roots of $x^\ell - 1 = \prod_i (x - \zeta^i)$ read modulo p, and $x^\ell - 1$ is separable over $\mathbb{F}_p$ because its derivative $\ell x^{\ell-1}$ does not vanish at a root, as $p \neq \ell$. Hence $\bar{\zeta} \in \mathbb{F}_p^\times$ has order exactly ℓ, and $\ell \mid p - 1$.

(3) Show that every prime $p \geqslant 5$ with $p \nmid \ell\Delta$ that splits completely in L is ordinary for E, and conclude. Where did you use $\ell \geqslant 3$?

Solution. **The hypothesis $\ell \geqslant 3$ is used exactly at $\ell \mid 2$.** Suppose $p \geqslant 5$ splits completely in L, with $p \nmid \ell\Delta$, and that E_p is supersingular. By Problem 1.12(2), $\#E_p(\mathbb{F}_p) = p + 1$, so part (2) gives $\ell \mid p+1$ and $\ell \mid p-1$, hence $\ell \mid 2$, contradicting $\ell \geqslant 3$. So every such p is ordinary. By Theorem 3.7 there are infinitely many primes that split completely in L, and removing the

finitely many primes dividing $6\ell\Delta$ leaves infinitely many ordinary primes. For $\ell = 2$ the two congruences say only that p is odd, and the argument gives nothing.

(4) Take the curve $E\colon y^2 + y = x^3 - x^2 - 10x - 20$, of discriminant -11^5, and $\ell = 3$. By part (2), a prime $p \neq 3, 11$ that splits completely in $\mathbb{Q}(E[3])$ has all nine 3-torsion points of $E_p(\overline{\mathbb{F}}_p)$ rational. `Magma` lists the supersingular primes below 100 and the primes below 3000 with that property:

```
E := EllipticCurve([0, -1, 1, -10, -20]);          // y^2 + y = x^3 - x^2 - 10x - 20
Discriminant(E);                                    // -161051
good := [p : p in PrimesUpTo(3000) | p ne 11];
[p : p in good | p lt 100 and IsSupersingular(ChangeRing(E, GF(p)))];
// [ 2, 19, 29 ]
full := function(p, l)       // are all the l-torsion points of E_p rational?
  I := Invariants(AbelianGroup(ChangeRing(E, GF(p))));
  return #I eq 2 and I[1] mod l eq 0;
end function;
split := [p : p in good | p ne 3 and full(p, 3)];  split;
// [ 337, 523, 1087, 2437, 2719, 2749 ]
[p mod 3 : p in split];                                    // [ 1, 1, 1, 1, 1, 1 ]
[IsSupersingular(ChangeRing(E, GF(p))) : p in split];
// [ false, false, false, false, false, false ]
[p : p in good | p lt 300 and p ne 5 and full(p, 5)];      // and the same for l = 5:
// [ 31, 41, 61, 71, 101, 131, 151, 181, 191, 211, 241, 251, 271, 281 ]
```

Check the output against parts (2) and (3), and compare the list of supersingular primes with the list in the example cited before this problem. The last line runs the same test for $\ell = 5$: which primes below 300 appear, and what does that suggest about the field $\mathbb{Q}(E[5])$? (The rational point $(5, 5)$ of E has order 5.)

Solution. **The good primes $p \equiv 1 \pmod 5$. That $\mathbb{Q}(E[5]) = \mathbb{Q}(\mu_5)$.** The supersingular primes below 100 are 2, 19 and 29, as in [Sil09, Example V.4.6] once the errata add $p = 2$. The primes $p < 3000$ for which all nine 3-torsion points of E_p are rational are all $\equiv 1 \pmod 3$ and all ordinary, as parts (2) and (3) predict for the primes splitting completely in $\mathbb{Q}(E[3])$.
For $\ell = 5$ the list is every prime $p < 300$ with $p \nmid 5\Delta$ and $p \equiv 1 \pmod 5$ (the bad prime 11 is also $\equiv 1 \pmod 5$, but the listing leaves it out, as part (2) says nothing about it), which suggests that $\mathbb{Q}(E[5])$ is the field $\mathbb{Q}(\mu_5)$ of 5-th roots of unity: the primes splitting completely in $\mathbb{Q}(\mu_5)$ are exactly those with $p \equiv 1 \pmod 5$.

Remark. That is right for this curve. The rational point $(5, 5)$ generates a subgroup $\mathbb{Z}/5\mathbb{Z}$ of $E[5](\overline{\mathbb{Q}})$ on which the Galois group acts trivially, and pairing with it under the Weil pairing, which is nondegenerate and Galois invariant [Sil09, Proposition III.8.1(c),(d)], identifies the quotient $E[5](\overline{\mathbb{Q}})/\langle(5, 5)\rangle$ with $\mu_5(\overline{\mathbb{Q}})$, compatibly with the Galois action. That $\mathbb{Q}(E[5])$ is no larger than $\mathbb{Q}(\mu_5)$ is a further fact about this curve that we do not prove.
For $\ell = 3$, the primes found and the primes splitting completely in $\mathbb{Q}(E[3])$ are in fact the same: for $p \nmid 3\Delta$, the Frobenius at a prime of $\mathbb{Q}(E[3])$ above p acts on the 3-torsion points as F^1 acts on $E_p[3](\overline{\mathbb{F}}_p)$, so p splits completely if and only if $E_p[3](\overline{\mathbb{F}}_p)$ is rational. We do not prove this here.

Remark 3.9 (Supersingular primes)**.** The other direction is much harder. For the curve of Problem 3.8(4), D. H. Lehmer found exactly 27 odd supersingular primes below 31500 [Sil09, Example V.4.6]. For an elliptic curve over $\mathbb{Q}$ without complex multiplication, the supersingular primes have density 0 (Serre, Elkies), and yet there are infinitely many of them (Elkies) [Sil09, Theorems V.4.7 and V.4.9]. The Lang–Trotter conjecture predicts that about $c\sqrt{x}/\log x$ of them lie

below x [Sil09, Conjecture V.4.8]. For a curve with complex multiplication, such as $y^2 = x^3 + x$, the supersingular primes have density 1/2 [Sil09, Examples V.4.4 and V.4.5].

References

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